

Farm Animal Demonstration

Title: Farm Animal Details using Classes and Objects[cite: 1, 18]

Objective/Aim:

To write a Java program to create farm animal objects and display their name, place, food, and sound using a constructor and method[cite: 1, 18].

System Requirements :

Hardware

- Standard PC/Laptop[cite: 1]
- Intel Core i3/i5 processor or equivalent[cite: 1, 18]
- Minimum 4 GB / 8 GB RAM[cite: 1, 18]
- Minimum 500 MB free storage[cite: 1]
- Keyboard and Monitor[cite: 1]

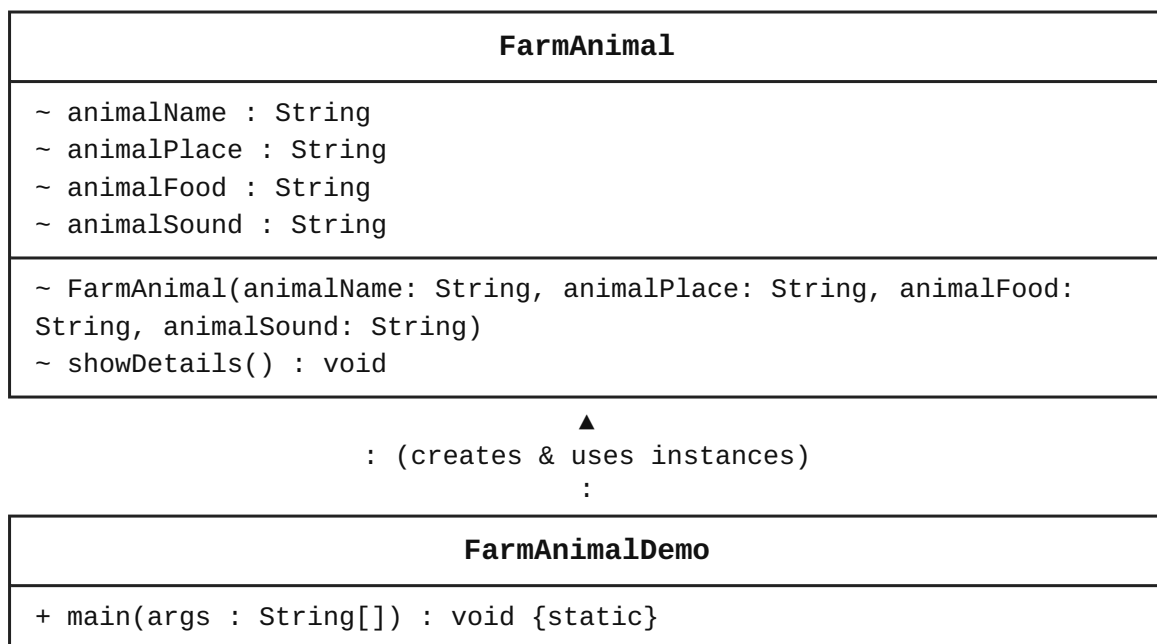
Software

- Operating System: Windows/Linux/macOS[cite: 1, 18]
- Java Development Kit (JDK 17 or JDK 21)[cite: 1, 18]
- Text Editor or IDE such as Eclipse, IntelliJ IDEA, NetBeans, or VS Code[cite: 1, 18]
- Command Prompt/Terminal[cite: 1]

Algorithm / Logic Description:

1. Create a class named `FarmAnimal` with data members `animalName`, `animalPlace`, `animalFood`, and `animalSound`[cite: 18].
2. Define a parameterized constructor to initialize all animal fields[cite: 18].
3. Define a `showDetails()` method to print the animal details in a structured format[cite: 18].
4. Create the `FarmAnimalDemo` class containing the `main()` method[cite: 1, 19].
5. Instantiate three `FarmAnimal` objects:
 - `cow`: ("Cow", "Cowshed", "Grass", "Moo")[cite: 1, 19]
 - `pig`: ("Pig", "Pigsty", "Vegetables", "Oink")[cite: 2, 19]
 - `horse`: ("Horse", "Stable", "Grass", "Neigh")[cite: 2, 19]
6. Call `showDetails()` on each animal object to display their respective details[cite: 18, 19].
7. Stop the program[cite: 2].

UML class Diagram



Relationship: FarmAnimalDemo creates objects (cow, pig, horse) and uses instances of the FarmAnimal class[cite: 3, 18].

Source Code:

```
class FarmAnimal {
    String animalName;
    String animalPlace;
    String animalFood;
    String animalSound;

    FarmAnimal(String animalName, String animalPlace, String animalFood, String animalSound) {
        this.animalName = animalName;
        this.animalPlace = animalPlace;
        this.animalFood = animalFood;
        this.animalSound = animalSound;
    }

    void showDetails() {
        System.out.println("Animal Name : " + animalName);
        System.out.println("Lives At : " + animalPlace);
        System.out.println("Food : " + animalFood);
        System.out.println("Sound : " + animalSound);
        System.out.println();
    }
}

public class FarmAnimalDemo {
    public static void main(String[] args) {
        FarmAnimal cow = new FarmAnimal("Cow", "Cowshed", "Grass", "Moo");

        FarmAnimal pig = new FarmAnimal("Pig", "Pigsty", "Vegetables", "Oink");
    }
}
```

```
        FarmAnimal horse = new FarmAnimal("Horse", "Stable", "Grass", "Neigh");

        cow.showDetails();
        pig.showDetails();
        horse.showDetails();
    }
}
```

Compilation & Execution Commands:

Save file as: FarmAnimalDemo.java[cite: 5, 19]

Compilation: javac FarmAnimalDemo.java[cite: 5, 19]

Execution: java FarmAnimalDemo[cite: 5, 19]

Output Screenshot:



```
PS C:\Users\mmeen\RAVI> & 'C:\Program Files\Java\jdk-26.0.1\bin\java.exe' '--enable-preview' '-XX:+S
howCodeDetailsInExceptionMessages' '-cp' 'C:\Users\mmeen\AppData\Roaming\Code\User\workspaceStorage\b
53c69995fa1d8f837d3efeb869e4899\redhat.java\jdt_ws\RAVI_49fdb1a2\bin' 'FarmAnimalDemo'
Animal Name : Cow
Lives At    : Cowshed
Food        : Grass
Sound       : Moo

Animal Name : Pig
Lives At    : Pigsty
Food        : Vegetables
Sound       : Oink

Animal Name : Horse
Lives At    : Stable
Food        : Grass
Sound       : Neigh

PS C:\Users\mmeen\RAVI>
```

Result:

The Farm Animal Details program was successfully implemented in Java using classes and objects[cite: 5]. The program demonstrates the use of a constructor, object creation, data members, and methods to store and display the details of various farm animals[cite: 5].